Spotting the Fakes: A Deep Dive into GAN-Generated Face Detection

XIN WANG and TING YU TSAI, University at Albany, State University of New York, Albany, New York, USA
LI LIN, Purdue University, West Lafayette, Indiana, USA
HUI GUO, University at Buffalo, State University of New York, Buffalo, New York, USA
SHU HU, Purdue University, West Lafayette, Indiana, USA
MING-CHING CHANG and PRADEEP K. ATREY, University at Albany, State University of New York, Albany, New York, USA
SIWEI LYU, University at Buffalo, State University of New York, Buffalo, New York, USA

Generative Adversarial Networks (GANs) have enabled the creation of highly authentic facial images, which are increasingly used in deceptive social media profiles and other forms of disinformation, resulting in serious consequences. Significant progress has been made in developing GAN-generated face detection systems to identify these fake images. This study offers a comprehensive review of recent advancements in GAN-generated face detection, focusing on techniques that detect facial images generated by GAN models. We categorize detection methods into three groups: (1) deep learning-based approaches, (2) physics-based methods, and (3) physiology-based methods. We summarize key concepts in each category, connecting them to relevant implementations, datasets, and evaluation metrics. Additionally, we provide a comparative analysis between automated detection and human visual performance to highlight the strengths and weaknesses of both approaches. Furthermore, we review related surveys, including detecting morphed faces, manipulated faces, DeepFake, and faces generated by diffusion models. Finally, we discuss unresolved challenges and suggest potential directions for future research.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; **Machine learning**; • **Applied computing** → **Computer forensics**; • **Security and privacy** → **Human and societal aspects of security and privacy**;

Additional Key Words and Phrases: Media forensics, GAN-generated face detection, disinformation, generative adversarial network, face synthesis, diffusion models, DeepFake, human visual performance

This work is partially supported by the National Science Foundation under Grant Nos. SaTC-2153112 (PI: S. Lyu) and IIS-2434967 (PI: S. Hu), Purdue University Seed Funding for High-Impact Review Papers (PI: S. Hu), and startup funds from the University at Albany (for X. Wang and P. K. Atrey).
Authors' Contact Information: Xin Wang, University at Albany, State University of New York, Albany, New York, USA; e-mail: xwang56@albany.edu; Ting Yu Tsai, University at Albany, State University of New York, Albany, New York, USA; e-mail: ttsai2@albany.edu; Li Lin, Purdue University, West Lafayette, Indiana, USA; e-mail: lin1785@purdue.edu; Hui Guo, University at Buffalo, State University of New York, Buffalo, New York, USA; e-mail: hguo8@buffalo.edu; Shu Hu, Purdue University, West Lafayette, Indiana, USA; e-mail: hu968@purdue.edu; Ming-Ching Chang, University at Albany, State University of New York, Albany, New York, USA; e-mail: mchang2@albany.edu; Pradeep K. Atrey (corresponding author), University at Albany, State University of New York, Albany, New York, USA; e-mail: patrey@albany.edu; Siwei Lyu, University at Buffalo, State University of New York, Buffalo, New York, USA; e-mail: siweilyu@buffalo.edu.

ACM 1551-6865/2025/7-ART193
https://doi.org/10.1145/3742786

ACM Reference format:
Xin Wang, Ting Yu Tsai, Li Lin, Hui Guo, Shu Hu, Ming-Ching Chang, Pradeep K. Atrey, and Siwei Lyu. 2025. Spotting the Fakes: A Deep Dive into GAN-Generated Face Detection. *ACM Trans. Multimedia Comput. Commun. Appl.* 21, 7, Article 193 (July 2025), 24 pages.
https://doi.org/10.1145/3742786

1 Introduction

The development of **Generative Adversarial Networks (GANs)** [49] has facilitated the creation of highly realistic human face images that are often indistinguishable from real ones [74, 76, 77]. These AI synthesized face images mark a significant milestone in AI Generated Content [16]. These GAN-generated faces, or GAN faces, can easily be used to create fake social media profiles and fraudulent accounts [57, 120, 121, 148], which can be exploited for malicious purposes, raising significant societal concerns. In a real-world instance, a high school student used a GAN-generated face to create a fictitious candidate during a voting event [120]. This deception fooled X (formerly known as Twitter) into awarding the fake candidate a blue checkmark, falsely verifying their legitimacy. Having bypassed verification, this false candidate could establish donation channels to siphon public funds, compromising the integrity of the election and undermining property-related laws. Fake social media accounts that use GAN faces also have serious social consequences [86, 108, 121]. For example, if such fraudulent accounts were linked to senior executives in an organization and made statements about its financial status, it could disrupt the stock market. Their deceptive information could mislead investors into making poor financial decisions, leading to substantial losses.

The growing need to automatically detect GAN-generated faces has spurred the development of various detection methods to combat the malicious use of these AI-synthesized faces [115, 141, 165]. However, detecting GAN faces remains complex and challenging, primarily due to two major obstacles. First, an effective GAN face detection method must be accurate and adaptable, capable of identifying a wide variety of GAN-generated images from different models while also being resilient to adversarial attacks. Second, the detection process and results should be *understandable to human users*, especially non-AI experts, rather than relying solely on complex **Deep Learning (DL)** models tailored to specific datasets. While automated techniques continue to advance, human visual performance in distinguishing GAN-generated faces remains a significant challenge. Research indicates that human observers frequently encounter difficulty in recognizing GAN faces, with accuracy levels that are comparable to random guessing (50%) [114]. Despite receiving training and feedback, their capacity to distinguish between real and synthetic features does not significantly improve [117]. These results underscore the importance of automated GAN face detection methods that can surpass human perception and deliver dependable, illuminating results.

In this article, we present a survey explicitly focused on detecting *GAN-generated faces*, which are *entirely AI-synthesized face images*. Such images have distinct properties compared to manipulated or morphed images since they are generated from scratch. The rapid proliferation of generative AI technologies [39] underscores the importance of effective detection methods to prevent misuse. Early approaches to GAN face detection primarily relied on DL techniques, achieving high accuracy in practice but often lacking interpretability. To address these limitations, recent research has shifted toward explainable methods, leveraging physical signals or physiological cues. For instance, some studies differentiate GAN-generated faces by examining inaccuracies in capturing authentic human physiological traits, such as inconsistencies in corneal highlights or pupil shapes [53, 61]. Typically, these approaches identify subtle yet meaningful discrepancies arising from the synthesis process itself, although they often require specific conditions (e.g., frontal views or clearly visible reflections). Additionally, unlike typical AI classification problems, GAN face detection uniquely demonstrates a scenario in which human observers perform significantly worse (approximately 50%–60% accuracy [114]) compared to algorithmic methods, highlighting the necessity of developing robust automated detection systems.

Despite recent advancements, significant challenges remain in reliably identifying GAN-generated faces, particularly with evolving GAN architectures and increasingly realistic results. Consequently, continued research into advanced, explainable, and robust methods is essential to address current shortcomings. Our survey contributes an in-depth analysis of detection methodologies, specifically focusing on GAN-generated faces, providing valuable insights to enhance accuracy, reliability, and transparency in face-forensics applications.

We summarize the key contributions of this article as follows:

— This article provides the first comprehensive review of the diverse range of GAN face detection methods to the best of our knowledge. To systematically organize and summarize the extensive literature, we categorize these methods into four key groups: (1) DL-based methods, (2) Physics-based methods, (3) Physiology-based methods, and (4) Methods based on human visual performance. Additionally, we emphasize explainable approaches that enhance human understanding by providing interpretability in the decision-making process and outcomes.
— We analyze the critical role of recognizing GAN-generated faces in verifying social interactions and detecting potential security and privacy violations. This article delves into human visual performance, offering strategies for identifying the subtle non-genuine features produced by GANs.
— We discuss several challenges facing current state-of-the-art methods and suggest potential directions for future research.

This article extends our previous conference work [154] in the following aspects. These additions deepen the scholarly contributions of our previous work and offer a more nuanced understanding of the evolving AI face detection landscape.

(1) *Incorporation of recent developments*: We have updated our survey with recent advancements, providing the latest insights and methodologies, including new subtopics or emerging trends that were not covered in the original article.
(2) *Enhanced coverage in related works*: The related surveys in Section 3 include a wider range of topics, such as diffusion model face detection (Section 3.5), offering a more comprehensive context.
(3) *Additional details and deeper analysis*: Expand on methodologies, datasets, and applications with a more in-depth analysis, offering comprehensive insights and comparisons.
(4) *Introduction of a new discussion section*: Section 3.6 explores the motivations behind our research and addresses key factors influencing the direction and implications of our work.

The remainder of this article is structured as follows. In Section 2, we summarize mainstream methods for generating high-quality faces using GAN models and their evaluation metrics. Section 3 covers related surveys and highlights the major differences between our survey and others. Section 4 organizes existing literature on GAN face detection into different categories. In Section 5, we summarize standard datasets. In Section 6, we discuss promising future research directions for developing forensic algorithms. Finally, we present conclusions from this survey in Section 7.

2 Background

2.1 Foundations of GANs: Theoretical Perspectives

GANs [49] consist of a *generator* $G(z)$ and a *discriminator* $D(x)$, both modeled as neural networks. The generator maps a random noise vector z from a prior distribution $p_z(z)$ to generate synthetic samples, while the discriminator distinguishes real samples from generated ones. GAN training follows a two-player minimax game:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}(x)}\left[\log D(x)\right] + \mathbb{E}_{z\sim p_z(z)}\left[\log(1 - D(G(z)))\right]. \tag{1}$$

Here, $D(x)$ aims to maximize its ability to classify real samples correctly, while $G(z)$ minimizes the discriminator's success in distinguishing real from fake data. However, direct optimization of this objective often leads to issues like vanishing gradients, where G fails to learn if D becomes too strong. To mitigate this, an alternative loss function is used for the generator:

$$\min_G L_G = -\mathbb{E}_{z\sim p_z(z)}\left[\log D(G(z))\right]. \tag{2}$$

This encourages G to generate samples that D classifies as real with high confidence. The GAN training process alternates between updating D and G using gradient-based optimization techniques such as Stochastic Gradient Descent or Adam. Convergence is ideally achieved when $D(x)$ would be 0.5 for all x, meaning real and generated data are indistinguishable.

GANs have been widely used in various applications, particularly in image generation tasks, leading to significant advancements in producing realistic, high-quality images from datasets such as ImageNet [34], CelebA [95], Cityscapes [26], and others. Notable applications include: Super-Resolution and Image Synthesis: GANs are employed to enhance image resolution, producing more detailed and high-quality versions [82, 134]. They can create photorealistic images from scratch [13, 124]. Creative Design and Art: GANs offer innovative tools for artists and designers, enabling the generation of art, music, and design patterns [77, 80, 159].

2.2 The Highly Realistic GAN-Generated Faces

We provide an overview of the most popular GAN models used to generate high-quality faces, as they are the primary focus of most GAN-based face detection methods. The surveys of [64, 155] provide additional information on the different types of GANs. In the past 5 years, various GAN models (e.g., PG-GAN [74], BigGAN [13], StyleGAN [76], StyleGAN2 [77]) have been developed to synthesize and generate realistic and diverse face images. These GANs effectively encode complex semantic information in their intermediate features [8] and latent space [48, 65, 142], enabling them to create high-quality face images from random noise. They can also produce facial images with different attributes, such as backgrounds, expressions, ages, and viewing angles. However, manipulating the latent space is limited to images generated by GANs, as they lack inference functions or encoders for real images. GAN inversion methods can invert a given image back into the latent space of a pre-trained GAN model, allowing precise reconstruction by the GAN generator [155]. This approach bridges the gap between real and fake face images and can significantly enhance the quality of generated faces. As a result, GAN inversion is increasingly adopted in state-of-the-art models like StyleGAN2 [77], StyleGAN3 [75], InterFaceGAN [142], and Image2StyleGAN++ [1].

2.3 Evaluation of GAN Face Generation and Detection

In addition to traditional evaluation metrics, several new metrics have been specifically designed to assess GAN-based face generation performance. These metrics are essential for evaluating the quality, diversity, and fidelity of generated images, offering deeper insights into the models' capabilities beyond standard classification metrics. Table 1 summarizes these metrics. One key

Table 1. Comparison of GAN Evaluation Metrics

Metric	Purpose	Strengths	Weaknesses
FID [59]	Measures similarity between real and generated images using feature statistics from the Inception network	Captures distributional differences, robust to mode collapse	High bias, requires a large number of samples for stability
IS [138]	Evaluates the quality and diversity of generated images based on classifier predictions	Simple and widely used, correlates well with human judgment	Sensitive to model parameters, insensitive to mode collapse
KID [10]	Similar to FID but uses MMD with polynomial kernels	More unbiased than FID for small sample sizes	Requires kernel selection, less common than FID
PPL [76]	Measures smoothness and consistency of latent space transformations in GANs	Evaluates latent space structure, useful for style-based GANs	Not applicable to all GAN architectures, sensitive to hyperparameters

metric is the **Fréchet Inception Distance (FID)** [59], which measures the distance between the feature vectors of real and generated images extracted from a pre-trained Inception network. FID is calculated as:

$$\mathrm{FID} = ||\mu_r - \mu_g||^2 + \mathrm{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{\frac{1}{2}}), \tag{3}$$

where μ_r and μ_g are the means, and Σ_r and Σ_g are the covariance matrices of the real and generated data feature vectors, respectively. Lower FID scores indicate that the generated images more closely resemble the real ones in distribution.

Another important metric is the **Inception Score (IS)** [138], which evaluates the quality and diversity of generated images. The score is calculated as:

$$\mathrm{IS} = \exp(\mathbb{E}_x D_{KL}(p(y|x)||p(y))), \tag{4}$$

where $p(y|x)$ is the conditional probability of label y given image x, and $p(y)$ is the marginal probability over all images. A higher IS indicates that the generated images are both realistic and diverse across different classes. The Mode Score extends the IS to penalize models that generate low-diversity images. It compares the output distribution of generated images with the true class distribution in the dataset, ensuring that the GAN doesn't overlook less frequent modes in the data. The **Kernel Inception Distance (KID)** [10] is another metric similar to FID, but based on the squared **Maximum Mean Discrepancy (MMD)**. KID uses polynomial kernels to compare the distributions of real and generated images:

$$\mathrm{KID} = \mathrm{MMD}^2(p_{\mathrm{data}}, p_g), \tag{5}$$

where p_{data} and p_g are the distributions of real and generated images, respectively. Unlike FID, KID has an unbiased estimator, making it more robust in certain scenarios.

Finally, **Perceptual Path Length (PPL)** [76] measures the smoothness and consistency of a GAN's latent space. It calculates the perceptual difference between pairs of images generated from neighboring latent vectors. A lower PPL indicates a well-structured latent space, leading to more coherent and consistent image transformations.

These metrics collectively offer a comprehensive framework for evaluating GAN-based face generation methods, enabling researchers to assess various aspects of model performance beyond mere visual inspection. Additionally, since GAN face detection is usually treated as a binary classification problem, metrics such as Accuracy, Precision, Recall, ROC analysis, and AUC are commonly used [136]. These metrics evaluate GAN face detection models, ensuring a well-rounded understanding of their strengths and weaknesses across different scenarios.

Table 2. A Summary of Related Surveys with the Survey Presented in This Article

Category	Survey Papers	Key Remarks	Evaluation Metrics Discussed	Datasets Covered	GAN Face Centric?
Morphed-Face Detection	[137, 146]	Focuses on morphed face generation and detection methods	Limited	Few	No
Manipulated-Face Detection	[113, 145]	Surveys cover methods for detecting manipulated faces, not necessarily GAN-generated faces	Yes	Several	No
DeepFake Detection	[111]	Covers DL-based DeepFake detection methods in general	Yes	Some	No
	[2, 83, 88, 127, 129, 164]	Focuses on DeepFake benchmarks and datasets	Yes	Extensive	No
	[99, 145, 167]	Reviews general DeepFake detection methods	Yes	Some	No
	[78]	Focuses specifically on audio DeepFakes	No	Few	No
	[103]	Covers audio-visual DeepFake detection	Yes	Some	No
Diffusion Model Generated Face Detection	[89, 92, 127]	Discusses detection of content generated by diffusion models	Yes	Some	No
GAN Face Detection	[93]	Focuses on DL-based detection methods only	Limited	Few	Yes (but overlooks non-DL-based methods)
	[154]	Provides an earlier overview of GAN face detection methods	Yes	Yes	Yes
	This article	Extends previous work, providing a comprehensive and up-to-date review of GAN face detection methods with a strong emphasis on explainability, datasets, and future challenges	Yes	Extensive	**Yes**

3 Related Surveys

While this survey focuses on detecting GAN-based full-face synthesis, the task is closely related to other fake face detection challenges [41, 112]. To provide a more comprehensive overview, we also include additional surveys in the related domains. Table 2 summarizes these related survey papers.

3.1 GAN Face Detection

A related survey in [93] focuses solely on DL-based GAN face detection methods and overlooks non-DL approaches and interpretability issues, which are crucial for applying these techniques in real-world scenarios. Additionally, the survey by Kammoun et al. [73] primarily focuses on face generation using GANs rather than on detection.

3.2 Morphed-Face Detection

Morphed-face detection aims to identify images created by blending two or more faces [130], a task made challenging by sophisticated morphing techniques. Developing effective and precise detection methods is crucial to prevent misuses [140], particularly as research continues to advance in both generating and detecting morphed faces [137, 146]. While GAN-based methods can be used for both generating and detecting morphed faces [28, 33, 58, 139, 147], existing surveys on morphed-face detection [137, 146] are generally broad and do not specifically focus on GAN-based morphing. Furthermore, the motivations behind generating morphed faces often differ from those for GAN-generated faces, as morphed faces are typically used for identity theft and profile impersonation.

Fig. 1. Video conferencing DeepFakes detection [56]. *Left*: A video call participant is actively authenticated using live patterns displayed on the screen. *Right*: A real person's cornea reflects the pattern shown, while a real-time DeepFake cannot replicate this reflection.

3.3 Manipulated-Face Detection

Manipulated-face detection focuses on identifying images altered or tampered with by AI algorithms [125], a growing concern as AI-generated images are increasingly used for fraudulent activities such as face swapping [68] and facial manipulation [94]. Current surveys [113, 145] emphasize the need for reliable and robust detection methods to protect the integrity of digital media and prevent misuse. It is important to distinguish between face manipulation and GAN-based face generation—the former modifies existing images, while the latter creates entirely new faces from scratch.

3.4 DeepFakes Detection

DeepFake detection involves identifying artificially generated or altered media, such as images, videos, audio, and text, created using AI [111, 143]. Such media can be used to deceive viewers, spread misinformation, and generate fake news [9, 122, 126]. Given their potential for misuse, developing reliable detection methods is crucial for maintaining the authenticity and credibility of digital content, especially on social media platforms [32, 71, 106, 161]. While some surveys have broadly addressed DeepFake detection [71, 145], others provide different perspectives or focus on specific media types. For example, Khanjani et al. [78] focus specifically on audio DeepFakes, an area that has often been overlooked, while Masood et al. [103] cover both audio and visual modalities. Important benchmarks in the field include [2, 83, 88, 127, 129, 164]. The COVID-19 pandemic has led to increased reliance on video conferencing, prompting a rise in real-time DeepFakes used for impersonation attacks during live video calls [46, 85]. These real-time DeepFakes, illustrated in Figure 1, present novel challenges for detection algorithms, as traditional methods may not adequately handle the complexities of live streaming contexts. Despite recent advances [56], DeepFake detection in video conferencing scenarios remains relatively underexplored. While GAN face detection methods are frequently mentioned in related literature, their methodologies are rarely thoroughly examined [153]. Our survey aims to bridge this gap by providing an in-depth analysis of GAN-based face detection approaches, highlighting explainable properties that differentiate real faces from GAN-generated ones, and addressing broader detection challenges to enhance overall accuracy and reliability.

In summary, DeepFakes detection is a broad topic. Although GAN face detection is often included in related surveys, it may lack a detailed analysis of the methods used to detect GAN-generated faces. The in-depth analysis and discussion of the GAN face detection methods in our survey, such as analyzing the explainable features that differentiate real and fake faces, can help researchers develop more effective detection techniques. By further investigating the GAN face detection methods, we can better understand the challenges and limitations of DeepFakes detection and develop more accurate and reliable methods for detecting GAN-based DeepFakes.

3.5 Diffusion Model Generated Face Detection

Surveys like [89, 92, 127] discuss detecting content created by diffusion models, but none specifically focus on detecting faces generated by these models. For instance, Lin et al. [89] address detection across various types of content, including text, images, audio, and video from large language models and diffusion models. Liu et al. [92] discuss detection methods for different models, including VAEs, GANs, and diffusion models, while Pei et al. [127] center on DeepFake generation. This section focuses on surveying detection algorithms for faces generated by diffusion models [157]. Given the potential for the use of diffusion models to spread misinformation and their realistic image outputs [22], this focus is crucial. Recent research emphasizes the need for robust detection methods across different architectures and datasets. For example, a study comparing ProGAN and Latent Diffusion models on synthetic and real images highlights the importance of adaptable detection strategies. This study underscores the need for detectors that can handle diverse diffusion model outputs by analyzing frequency and local patches [27]. In various practical scenarios, the multilocal intrinsic dimensionality technique has achieved near-perfect detection results. Lorenz et al. [97] demonstrated that this lightweight method effectively distinguishes between diffusion-generated images and real ones, even though it was originally designed for detecting adversarial cases. **Vision Transformers (ViTs)** offer a new approach to detecting DeepFake images generated by models like Stable Diffusion and StyleGAN2. By leveraging ViTs' global feature extraction, researchers have achieved high detection accuracy, outperforming traditional CNN-based models and addressing DeepFake detection as a multiclass challenge [3]. These studies contribute to tackling the growing challenges of synthetic media by providing comprehensive evaluations and diverse methods for detecting faces generated by diffusion models [3, 27, 97].

We summarize recent research on diffusion face detection into three main categories. First, *Physiology-Based Methods* focus on distortions in fine details of human features such as eyes, teeth, and fingers, highlighting discrepancies commonly seen in DeepFake images [12]. Second, *DL-Based Methods* encompass several approaches: spatial feature-based methods, such as the Denoising Diffusion Probabilistic Mask by Yang et al. [160] for accurate differentiation of real and fake faces, Edge Graph Regularization by Cheng et al. [24] which integrates edge graphs to enhance precision, and the Stepwise Error for Diffusion-generated Image Detection by Ma et al. [98] utilizing error analysis for improved accuracy. Nguyen et al. [110] leveraged gradient-based features to distinguish between diffusion-generated and authentic images.

In addition, Kamat et al. [72] addressed spatial and temporal generalization challenges in DeepFake video detection with their Simulated Generalizability Evaluation method. Deng et al. [35] explored frequency-domain characteristics via Fourier and wavelet transforms, employing a multi-scale network for detailed feature analysis. For real-world applications, Porcile et al. [131] demonstrated effective semantic-level artifact detection, particularly in compressed profile photos. Additionally, Asnani et al. [4] proposed frameworks to estimate generative model architectures from images, and Ţânţaru et al. [168] developed a weakly supervised approach for localizing DeepFake regions. Third, *Human Visual Performance* studies by Papa et al. [123] indicate that trained models surpass non-expert human capabilities in recognizing fake faces, highlighting the need for advanced solutions against misinformation. In summary, GAN-generated faces remain generally more consistent with real faces in geometric and photometric aspects compared to recent diffusion-generated faces [11].

3.6 Significance of Our Survey Relative to Existing Research

We have detailed the differences and advantages of this article compared to related surveys in Sections 3.1 to 3.5. In this subsection, we further address and elaborate on several key questions that emerge from the aspects highlighted in our survey, reinforcing the significance of our work.

(1) *Why focus only on GAN-generated faces?* GAN faces are particularly problematic because they are frequently used in misinformation and fraud. By focusing on GAN-generated faces, we address a major type of fake image and advance detection methods, which is crucial for improving AI security.
(2) *Why is explainability important for detection methods?* Explainability is crucial because it provides users, especially non-experts, with clear reasons for why a face is identified as fake. Instead of just relying on AI predictions, the detection method should offer concrete evidence, like irregular pupil shapes or other artifacts of a synthesized face. This transparency helps researchers understand how GAN faces differ from real ones, leading to more effective detection algorithms and better generation techniques.
(3) *Could simpler, explainable models give an advantage to adversaries if they are less accurate?* Combining explainable models with DL approaches balances transparency and accuracy. Explainable models offer insight and clarity, while DL models excel in performance. This hybrid approach ensures both high performance and interpretability [5], creating robust and trustworthy AI systems without compromising the benefits of DL.
(4) *Why is improving human performance important?* Though humans may not match the accuracy of detection models, they can notice subtle cues and inconsistencies, such as irregular pupil shapes, that automated systems might overlook. By enhancing human detection skills, we can better complement AI systems, strengthening defenses against fake images and safeguarding individuals and society.

4 GAN-Generated Face Detection Methods

4.1 Taxonomy of GAN Face Detection

We categorize the literature on GAN-generated face detection into four main groups based on their objectives and core principles, as illustrated in Figure 2, though there may be overlaps between categories. Based on the taxonomy, we review the existing GAN face detection methods systematically. Table 3 summarizes popular GAN face detection algorithms, detailing the datasets used and comparing their performance.

4.2 DL-Based Methods

DL techniques using GANs focus on detecting fake faces by extracting features at the signal level and training **Deep Neural Network (DNN)** classifiers in an end-to-end framework [14, 42, 67]. The initial work of Do et al. [36] utilized VGG-Net [144] to detect GAN-generated faces, which are mostly generated using DC-GANs [134] and PG-GAN [74]. VGG-16 architecture is employed with pre-trained weights of VGG-Face [15]. Real faces are collected from the CelebA face dataset [95] to train these networks. Mo et al. [107] found that signals in the residual field can be effective for distinguishing between real and GAN faces. They used high-pass filters on the input faces and then processed the residuals with deep networks for GAN face detection. Li et al. [84] focused on chrominance color components, extracting features to obtain color image statistics, and used these features to train a GAN face classifier. Similarly, Chen et al. [20] found that both luminance and chrominance signals enhance GAN face detection. In a more recent study, Fu et al. [43] used a dual-channel CNN to address the effects of common image post-processing operations. Their network

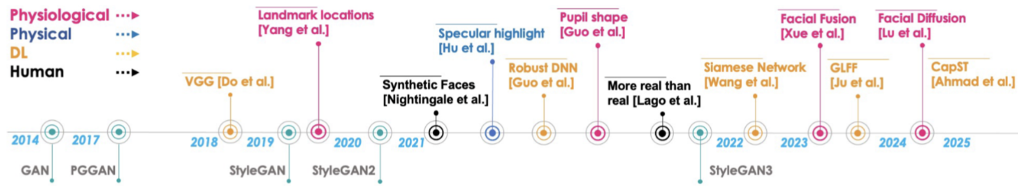

Fig. 2. A brief chronology of GAN face generation and detection works: *Generation*: The first GAN model, introduced in 2014, could only generate 32 × 32 pixel faces. After 2017, StyleGAN models emerged, capable of producing highly realistic faces that are difficult to distinguish from real ones. *Detection*: The earliest detection methods, primarily based on DNNs, appeared in 2018. However, due to limitations in performance and interpretability, techniques utilizing physical and physiological cues were developed between 2019 and 2021. Since StyleGAN2 faces are particularly challenging to detect, research into human visual performance on GAN-generated faces has been ongoing since 2021. These methods mark significant milestones and breakthroughs in the field.

Table 3. Summary of GAN Face Detection Methods, Including the Datasets, Statistics, and Performance Scores

Paper	Category	Method	Real Face (#Test)	GAN Face (#Test)	Performance
Nightingale et al. [114]	Human	Visual	FFHQ (400)	StyleGAN2 (400)	Acc: 0.5–0.6
Lago et al. [81]	Human	Visual	FFHQ (150)	StyleGAN2, etc. (150)	Acc: 0.26–0.8
Do et al. [36]	DL	CNN	CelebA (200)	PGGAN, DCGAN (200)	Acc: 0.80
Dang et al. [29]	DL	CNN	CelebA (1,250)	PCGAN (1,250)	Acc: 0.98
Mo et al. [107]	DL	CNN	CelebA-HQ (15K)	PGGAN (15K)	Acc: 0.99
Nataraj et al. [109]	DL	CNN	CelebA (500)	StarGAN (4,498)	Acc: 0.99
Fu et al. [43]	DL	CNN	CelebA-HQ (7K)	PGGAN (7K)	Acc: 0.98
Marra et al. [102]	DL	Incremental Classifier	-	StarGAN (2.4K), etc.	Acc: 0.815–1
Mansourifar et al. [100]	DL	Out of Context Object Detection	-	StyleGAN (100)	Acc: 0.80
Wang et al. [151]	DL	DNN	FFHQ (1K)	StyleGAN2 (1K), etc.	Acc: 0.88–0.991
Li et al. [84]	DL	Disparities in Color Components	CelebA-HQ, FFHQ (50K)	StyGAN, ProGAN (50K)	Acc: 0.997
Wang et al. [152]	DL	CNN	FFHQ (1K)	StyleGAN (1K)	Acc: 0.84
Hulzebosch et al. [63]	DL	ForensicTransfer	FFHQ (3K), etc.	StyleGAN (3K), ProGAN (3K), etc.	Acc: 0.01–1
Goebel et al. [47]	DL	CNN	CelebA (164), CelebA-HQ (1.5K)	StarGAN (1,476), ProGAN (3.7K)	Acc: 0.6768–0.849
Liu et al. [96]	DL	CNN	FFHQ, CelebA-HQ (10K)	StyleGAN (10K), PGGAN (10K), etc.	Acc: 0.9854–0.991
Chen et al. [19]	DL	Xception	FFHQ (7K)	LGGF (14K)	Acc: 0.99
Chen et al. [20]	DL	Improved Xception	CelebA (20,260)	PGGAN (20,260)	Acc: 0.713–0.977
Gragnaniello et al. [51]	DL	CNN	RAISE (≤7.8K)	StyleGAN2 (3K), ProGAN (3K), etc.	Acc: 0.928–0.999
Chen et al. [21]	DL	CNN	FFHQ (20K)	StyleGAN (20K), etc.	Acc: 0.9895–1
Guo et al. [55]	DL	Residual Attention	FFHQ (748)	StyleGAN2 (750)	AUC: 1
Nowroozi et al. [118]	DL	Cross-Co-Net	FFHQ (4K)	StyleGAN2 (4K)	Acc: 0.998
Dong et al. [37]	DL	Enhanced Spectrum-Based CNN	FFHQ (3.2K)	StyleGAN, StyleGAN2 (3.2K)	Acc: 0.95–0.96
Wang et al. [149]	DL	Siamese Network	FFHQ, CelebA-HQ (10K)	ProGAN, StyleGAN3 (20K)	AUC: 0.996–1
Hu et al. [61]	Physical	Corneal Specular Highlight	FFHQ (500)	StyleGAN2 (500)	AUC: 0.94
Matern et al. [104]	Physiology	Eye Color	CelebA (1K)	ProGAN (1K), Glow (1K)	AUC: 0.70–0.85
Yang et al. [162]	Physiology	Landmark Locations	CelebA (≥50K)	PGGAN (25K)	AUC: 0.9121–0.9413
Guo et al. [53]	Physiology	Irregular Pupil Shape	FFHQ (1.6K)	StyleGAN2 (1.6K)	AUC: 0.91

Note that the datasets used are often self-collected and may differ across papers, so performance scores should not be viewed as directly comparable.

features a deep CNN for pre-processed images and a shallow CNN for high-frequency components of the original image.

Real-World Applications of GAN Face Detection. A framework for evaluating detection methods across different models, datasets, and post-processing techniques was developed by Hulzebosch et al. [63], which this framework assesses features produced by common image pre-processing methods. Recent research has explored various feature-based models [21, 47, 96, 152]. However, the results from these models often lack interpretability, making it unclear why certain decisions are made for specific input faces.

Advanced, Incremental, and One-Shot Learning. Mansourifar et al. [100] recently investigated a one-shot GAN face detection method that uses scene understanding to identify out-of-context objects in GAN faces, distinguishing them from real ones. Additionally, Marra et al. [102] explored incremental learning techniques to detect and classify new GAN-generated faces while preserving performance on previously learned capabilities. Recent studies [37, 51, 66, 151, 152] provide additional difficulty analysis and systematic evaluations using advanced DNNs for GAN face detection. These works examine both visible and hidden anomalies. For instance, Wang et al. [152] found that CNN-generated images often exhibit common systematic flaws, making them relatively easy to identify. Additionally, Gragnaniello et al. [51] evaluated existing GAN face detection methods across various datasets and metrics, revealing that we are still far from achieving reliable detection of GAN-generated images. Unfortunately, the results from the aforementioned methods discussed in this subsection remain unclear. To address this issue, Guo et al. [55] introduced an attention-based approach to identify GAN-generated portraits by focusing on inconsistencies in the eyes. This model successfully detected anomalies in the iris by comparing the attention maps of GAN-generated faces with real ones, revealing noticeable differences. Specifically, the attention maps highlighted artifacts on the irises of GAN faces, while real faces showed no significant anomalies. However, the attention map alone does not fully explain the model's decision-making process.

A categorization of DL-based GAN face detection methods is presented in Table 4. In summary, while DL-based methods for GAN face detection have shown impressive performance [52, 166], understanding the model's decision-making remains challenging. Explainability is crucial for real-world applications where users need to trust the AI's reasoning, such as in scenarios where people question the authenticity of images and seek clear explanations for the AI's conclusions.

4.3 Physics-Based Methods

Physics-based methods detect GAN-generated faces by identifying artifacts or discrepancies related to real-world physical characteristics, such as lighting and reflections. Early work by Johnson et al. [69] explored how internal camera characteristics and light source directions could reveal image manipulation through distortions in the specular highlights of the eyes. Building on this, Matern et al. [104] observed that early GAN face variants [74] often exhibited unrealistic specular reflections, such as simple white blobs, though these issues have been largely resolved in newer models like StyleGAN2. Hu et al. [61] further advanced this approach by identifying GAN faces through inconsistencies in specular highlights between both eyes, using pixel-wise **Intersection of Union (IoU)** to detect discrepancies. As illustrated in their work, genuine human eyes typically have similar specular highlights, whereas GAN-generated eyes may show variations in the number, shape, or placement of these highlights. However, this method relies heavily on assumptions about frontal eye posture, lighting conditions, and the presence of specular highlights, which can lead to false positives when these assumptions are not met. Despite these limitations, physics-based detection techniques are generally more robust against adversarial attacks and provide interpretable results for human users [40, 61].

4.4 Physiology-Based Methods

Physiology-based methods examine the semantic aspects of human faces, such as symmetry, iris color, and pupil shapes, to identify artifacts that expose GAN-generated faces [25, 158]. Early studies [101, 105, 163] revealed that faces generated by StyleGAN [74] often exhibit noticeable anomalies, like asymmetry and inconsistent iris colors [104]. Yang et al. [162] found that while GANs can generate realistic facial components (e.g., eyes, nose, skin, mouth), they do not always ensure these

Table 4. Categorization of DL-Based GAN Face Detection Methods

Category	Methods	Key Features
CNN-Based Approaches	Standard and Xception-Based CNN Detectors	Utilizes CNN architectures, including XceptionNet variations, for feature extraction and classification [19, 29, 36, 107]
	Enhanced CNN-Based Methods	Combines CNNs with domain adaptation, spectral analysis, or multi-scale patch analysis for improved detection [20, 63, 152]
	Dual-Channel CNN	Uses a dual-channel approach to enhance feature representation for robust GAN face detection [43]
Attention-Based Methods	Residual and Self-Attention Networks	Uses residual attention and self-attention mechanisms to enhance feature learning [55, 118]
Frequency-Based Analysis	Fourier and Wavelet Transform-Based Detectors	Uses Fourier Spectrum and Discrete Wavelet Transform for analyzing GAN-specific artifacts [35, 37, 109]
	Color Component Disparities and Transfer Learning-Based Detectors	Analyzes color inconsistencies in GAN-generated faces and adapts pre-trained models for cross-domain DeepFake detection [47, 84]
	Global Texture Enhancement	Leverages texture-based enhancements for fake face detection in real-world settings [96]
Patch-Based Detection	Siamese and Patch-Based Networks	Uses Siamese networks and patch-level analysis to detect inconsistencies in GAN-generated images [19, 149]
	Global and Local Feature Fusion	Combines global and local features for improved GAN detection [21]
Anomaly Detection Approaches	Autoencoder and Feature Fusion-Based Methods	Utilizes autoencoders and deep networks integrating spatial and frequency features to detect inconsistencies [51, 151]
Few-Shot and Self-Supervised Learning	Incremental Learning and Object-Level Inconsistency Detection	Uses incremental classifiers and object anomaly detection to improve robustness against new GAN models [100, 102]

features are coherently or naturally placed on the face. They suggested using facial landmarks (e.g., the tips of the eyes, nose, and mouth) to detect these inconsistencies, as these landmarks can be automatically identified and used to train a classifier to distinguish GAN faces.

However, GAN face generation has advanced significantly. The quality of images produced by StyleGAN2 has improved, eliminating many of the earlier physiological artifacts [74, 76, 77]. StyleGAN3 [75] further enhances face synthesis by offering a more natural hierarchy of feature scales and improved physiological consistency, making the generated features more robust to translation and rotation.

A newer physiology-based GAN face detection method, introduced in [53], is motivated by the observation that GAN-generated faces often exhibit irregular pupil shapes. While genuine human pupils typically form smooth circles or ellipses, GAN-generated faces may display irregular or distorted shapes. This artifact is consistent across all known GAN models to date, present in both the synthesized human and animal images, including PG-GAN [74], StyleGAN3 [75], and SofGAN [18]. The presence of such artifacts is primarily due to GANs' lack of understanding of human eye anatomy, particularly the geometry and shape of the pupils. The GAN face detection method of [53] involves first localizing the eyes and then segmenting the pupil regions. The pupil boundary is then fitted with an ellipse model, and the "circularity" of the pupils is estimated by calculating a boundary-sensitive IoU [23] between the extracted pupil mask and the fitted ellipse. However, this approach can still result in false positives in cases where real faces have non-elliptical pupils due to conditions like diseases or infections. In summary, physiology-based methods are more interpretable, but they are still limited by environmental factors such as eye visibility and occlusion, similar to other forensic techniques.

4.5 Human Visual Performance

While many automated GAN face recognition techniques have been developed, the ability of humans to identify GAN-generated faces has not been extensively studied. Detecting GAN faces is more challenging for human eyes than typical image recognition. Therefore, it is crucial to explore human capabilities in recognizing GAN faces, considering the ethical and social implications [7]. Precision-recall and ROC curves are often used to evaluate automated GAN face recognition algorithms. While these metrics can be used to study human perceptual performance, they do not fully capture how realistic GAN faces can mislead the public. Although human performance is biased, it can be enhanced by using subtle but relevant cues, such as physiological indicators [79]. Earlier research [81] examined the human ability to recognize GAN-generated faces using a dataset of 150 GAN faces and 150 real faces. The GAN faces were generated by advanced models like PG-GAN, StyleGAN, and StyleGAN2, while the real faces were selected from the **Flickr-Faces-HQ (FFHQ)** dataset. In the study, 630 participants were asked to differentiate between 30 faces in 34 sequential challenges, with features chosen randomly in equal numbers from each category. The researchers found that participants struggled to identify the newer GAN faces accurately, and neither the speed of the test nor prior exposure to synthetic faces significantly impacted their performance.

In an early study [81], the human capacity to identify fake features was assessed. The dataset comprises 150 GAN faces and 150 actual faces. GAN profiles are generated from state-of-the-art GANs, including PG-GAN, StyleGAN, and StyleGAN2, and real profiles are selected from the FFHQ dataset. The 630 participants conducted 34 tasks in a sequential manner, each of which required them to distinguish 30 faces. The features were randomly selected from each category in equal proportions. The results indicated that the participants had lost the capacity to evaluate the newer GAN faces. The accuracy of the test is not affected by the test's pace or the fact that the participants have previously encountered synthetic faces produced by the generators. A recent study [114] explored how well individuals can distinguish GAN-generated faces from real ones. The study used 400 StyleGAN2 faces and 400 FFHQ faces, ensuring diversity in gender, age, and color across two test sets. Initially, 315 participants achieved around 50% accuracy when shown a limited number of GAN and real faces. In a second round, 170 new participants received a tutorial on identifying GAN artifacts and were given feedback after the event. However, this training only slightly improved the average accuracy. The study concluded that StyleGAN2 faces are realistic enough to fool both naive and trained observers. Further research on this topic can be found in [117]. In a third experiment, the study examined whether synthetic faces evoke the same level of trustworthiness as real faces, which could also aid in distinguishing GAN faces. However, the study did not specify the synthesis artifacts used for participant training. It suggests that the human ability to detect GAN faces can be enhanced with cues like physiological indicators (e.g., pupil shapes [53]) and dataset biases (e.g., GAN faces trained with biased FFHQ samples).

As GAN technology advances, distinguishing between real and synthetic faces will become increasingly difficult. Identifying general and consistent markers that help human eyes detect GAN faces is crucial. Indicators of AI-manipulated faces, such as morphing, swapping, and painting, could be useful. An open platform described in [54] tests the human ability to distinguish AI-generated faces from real ones using these cues, which can improve GAN face synthesis by making features harder to detect. In conclusion, studying human visual performance is essential for developing detection techniques and understanding the shortcomings of GAN-generated faces more comprehensively.

Table 5. Comparison of Datasets

Dataset	Images	Resolution	Annotations	Diversity	Limitations
FFHQ [76]	70,000	1,024 × 1,024	Facial landmarks	High (global variety)	Limited to faces
CelebA [95]	200,000+	178 × 218	40 attributes annotations	Medium (celebrity bias)	Celebrity-centric dataset
CelebA-HQ [74]	30,000	1,024 × 1,024	Same as CelebA	Medium (celebrity bias)	Limited to CelebA examples
RAISE [30]	8,000+	High-res (varies)	None	N/A	Not specific to faces
AI-Face [91]	1,646,545	Varies	Demographic annotations	High (global variety)	Limited to faces

5 Datasets

The rapid advancement of AI in both discriminative and generative models has led to the creation of numerous human facial datasets. Real face images are primarily sourced from datasets like FFHQ [76], CelebA [95], CelebA-HQ [74], RAISE [30], and AI-Face [91] among the others. Table 5 provides a detailed comparison of these datasets, highlighting their number of images, resolution, annotations, diversity, and limitations.

FFHQ [76]: The FFHQ dataset comprises 70,000 images at a resolution of 1,024 × 1,024 pixels. These images feature a wide range of ages, countries, and backgrounds, adding to the dataset's diversity and unpredictability. FFHQ includes high-quality facial images with various hairstyles, glasses, hats, and lighting conditions, making it an excellent resource for GAN training. FFHQ is developed by NVIDIA for StyleGAN research, and its high resolution and diversity have made it a standard for evaluating face generation models.

CelebA [95]: CelebA contains over 200,000 celebrity images, each annotated with 40 facial attribute parameters. The dataset features a variety of poses, facial expressions, lighting conditions, and occlusions, making it valuable for both discriminative and generative modeling. CelebA is widely used in tasks like facial attribute prediction, face detection, and face synthesis, and it serves as a benchmark in computer vision.

CelebA-HQ [74]: CelebA-HQ is an enhanced version of the CelebA dataset, consisting of 30,000 high-quality images. These images have been processed to achieve higher resolution (1,024 × 1,024) with fewer compression artifacts. CelebA-HQ is frequently utilized in GAN research, particularly for high-resolution image synthesis, such as in the development of PGGAN and StyleGAN.

RAISE [30]: The Raw Images Dataset (RAISE) comprises nearly 8,000 high-resolution, unedited DSLR photos, primarily used for image forensics and compression studies. While not specifically a face dataset, RAISE is valuable for evaluating real-world image processing methods and is frequently used with GANs for super-resolution and image synthesis tasks. State-of-the-art GAN models, such as those described in [19], enable advanced facial image synthesis. Newer GAN face datasets typically use images generated by StyleGAN2, while older ones rely on PGGAN images. NVIDIA has also developed a StyleGAN3 dataset specifically for evaluating GAN face detection performance. Table 3 lists popular GAN face detection datasets, with each study often utilizing self-collected datasets comprising different subsets tailored to specific techniques. For example, the work in [53] focuses on face photographs with visible pupils for training and evaluation. Despite advancements in the field, there is still a need for a large-scale benchmark dataset to facilitate the empirical evaluation of GAN face identification methods.

AI-Face [91]: The AI-Face dataset is the first million-scale demographically annotated AI-generated face image dataset, containing 1,646,545 face images. It includes real faces, faces from DeepFake videos, and faces generated by GANs and Diffusion Models. Each instance consists of an image associated with demographic annotations (gender, age, skin tone) and a target label (fake or real). This dataset serves as a critical resource for developing and evaluating GAN face detectors, particularly in addressing subpopulation shift problems—where detection performance varies across demographic groups, potentially leading to unfair targeting or exclusion of specific populations, an issue highlighted in the development of reliable GAN face detectors [89]. For instance, studies by Ju et al. [70] and Lin et al. [90] leverage demographically annotated datasets to tackle these disparities in AI-generated face detection. By providing a comprehensive and diverse set of annotated images, this dataset will facilitate the advancement of fairer and more robust AI face detection technologies.

6 Future Research Directions

Following our detailed review of GAN-based face detection methods, including recent advancements, various methodologies, and comparisons of models, training strategies, and metrics, we now explore promising future research directions. These directions aim to advance the development of more effective, interpretable, robust, and adaptable forensic algorithms.

6.1 Competing with Rapidly Evolving GAN Models

Despite their current limitations, GAN models are improving, and new, more powerful versions are expected soon. To address issues like irregular pupil shapes [53], inconsistent corneal highlights [61], and symmetry inconsistencies (e.g., earrings), we propose adding relevant constraints to existing models. However, the optimal way to enforce these constraints remains unclear. Enhanced neural network architectures, improved training methods [44], and larger datasets are needed to advance GAN models. StyleGAN3 [75], for example, has revised all signal processing components from StyleGAN2 to enhance the texture and 3D modeling of GAN-generated features. This underscores the increasing need for better GAN face detection methods and effective cues for identifying these faces.

Low-Power Demands. Efficient GAN face detectors that work on edge devices are also important. Forensic tools should be able to operate on smartphones, given the direct impact of GAN faces on identities and social networks. Research is needed to transition high-performance GPU models to mobile applications effectively.

6.2 Development of Interpretable Methods

Many GAN face detection systems lack interpretability. Current attention mechanisms [55] only show which pixels the network focuses on without explaining why certain pixels enhance performance. While physical and physiological methods [53, 61] offer more interpretable results, their assumptions (e.g., inconsistencies in iris or pupil) may not always apply to future GAN models. The optimal way to develop an interpretable end-to-end GAN face detection system using physical and physiological inputs remains uncertain.

Learning Multiple Cues. Our review indicates that methods relying on a single or few cues often struggle with performance, scalability, and interpretability in complex real-world scenarios like occlusions and chaotic data. Combining multiple attributes or artifacts from different cues is challenging. The goal is to enhance the generalization of detection systems and integrate various stimuli into a unified framework. To improve GAN face detection, consider using ensemble learning [135], multi-modal/task learning, and knowledge distillation [50].

6.3 Robust to Adversarial Attacks

DNNs are vulnerable to *adversarial attacks*, which exploit weaknesses in GAN face detection systems by introducing intentional disruptions or perturbations. While research often targets generic classifiers, the growing sophistication of adversarial techniques necessitates a focus on detecting fake faces specifically [60]. Studies on countering adversarial attacks include [17] and

[45], which address ways to evade detection by altering entire images, making these disturbances more noticeable.

Notably, the Key Region Attack [87] is a sparse attack strategy targeting critical pixels. This method localizes perturbations, making them harder to detect while focusing on specific areas of the image. Future GAN face detection algorithms must be designed with resilience to these sophisticated adversarial attacks in mind.

6.4 Handling Imbalanced Data Distribution

In online applications, authentic faces significantly outnumber GAN-generated ones, creating an imbalance in data distribution that poses challenges for GAN face detection methods trained on balanced datasets, as they often achieve high accuracy but low sensitivity in real-world scenarios where GAN faces are rare. To address this issue, techniques such as those proposed by Guo et al. [55] and Pu et al. [132] optimize the ROC-AUC by approximating and relaxing the AUC using Wilcoxon-Mann-Whitney statistics, thereby improving performance on imbalanced datasets; however, effectively handling highly uneven real-world data remains an open challenge.

6.5 Handling Mixtures with Other Face Manipulations

To keep pace with evolving face tampering technologies like Face Morphing [116], Face Swapping [128], and Diffusion-synthesized faces [11], GAN face detection forensics must be increasingly resilient. Beyond just *detection*, there is growing emphasis on *attribution* (identifying the tools and origins of GAN faces) and *characterization* (determining intent and potential harm). The DARPA Semantic Forensics (SemaFor) program [31] is one such initiative addressing these challenges.

6.6 Foundation Models for AI Face Detection

Foundation Models for AI face detection represent a significant advancement in the field of AI, specifically designed to identify increasingly sophisticated AI-generated faces [62]. These models leverage DL techniques, such as Contrastive Language Image Pre-Training [133], along with large datasets, to enhance detection accuracy. By strengthening face detection systems, they address critical issues related to digital security and content authenticity, making them essential tools in the fight against disinformation and digital fraud.

6.7 Agent for AI Face Detection

An AI agent is an intelligent system that autonomously performs tasks while adapting to different environments and constraints [156]. Unlike conventional DL models that operate as static classifiers, AI agents can dynamically adjust their decision-making strategies based on contextual information [150]. These agents leverage reinforcement learning, multimodal data fusion [6], and explainable AI techniques to enhance detection accuracy, robustness, and interoperability [38]. By incorporating an agent-based approach, AI face detection systems can become more efficient in handling challenges such as varying lighting conditions, occlusions, and adversarial attacks. Additionally, multi-agent architectures enable collaboration between different AI agents specialized in detection, recognition, and analysis, further improving overall performance and reliability [119]. Expanding beyond standalone detection, future developments can integrate multi-agent collaboration, where different AI agents specialize in face detection, recognition, emotion analysis, and anomaly detection.

7 Conclusion

This article reviews the latest advancements in GAN-based face detection algorithms, comparing state-of-the-art model architectures, training methods, and evaluation criteria, while highlighting the persistent challenges in detecting GAN-generated faces in real-world scenarios, which remain both difficult and highly sought after. Key considerations include computational efficiency, robustness, scalability, and addressing the ethical and societal impacts of combating misinformation to enhance digital trust. Developing advanced detection algorithms and establishing clear benchmarks are promising research directions, as this review underscores the interdisciplinary nature of GAN face detection, linking it to computer vision, machine learning, and cybersecurity, while also extending to related areas such as face morphing and face swapping. By synthesizing current knowledge and identifying areas for improvement, this study aims to inspire advancements in fake face detection technology, ultimately improving both accuracy and reliability.

Acknowledgment

The copyrights of the figures are retained by their respective authors and rights holders, with sources cited in the figure captions.

References

[1] Rameen Abdal, Yipeng Qin, and Peter Wonka. 2020. Image2StyleGAN++: How to edit the embedded images? In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 8293–8302.

[2] Enes Altuncu, Virginia N. L. Franqueira, and Shujun Li. 2024. Deepfake: Definitions, performance metrics and standards, datasets and benchmarks, and a meta-review. *Frontiers in Big Data* 7 (2024), 1400024.

[3] Muhammad Asad Arshed, Shahzad Mumtaz, Muhammad Ibrahim, Christine Dewi, Muhammad Tanveer, and Saeed Ahmed. 2024. Multiclass AI-generated deepfake face detection using patch-wise deep learning model. *Computers* 13, 1 (2024), 31.

[4] Vishal Asnani, Xi Yin, Tal Hassner, and Xiaoming Liu. 2023. Reverse engineering of generative models: Inferring model hyperparameters from generated images. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 12 (Dec. 2023), 15477–15493.

[5] Pranjal Atrey, Michael P. Brundage, Min Wu, and Sanghamitra Dutta. 2025. Demystifying the accuracy-interpretability trade-off: A case study of inferring ratings from reviews. arXiv:2503.07914. Retrieved from https://arxiv.org/abs/2503.07914

[6] Pradeep K. Atrey, Md Anwar Hossain, Abdulmotaleb El Saadik, and Mohan Kankanhalli. 2010. Multimodal fusion for multimedia analysis: A survey. *Multimedia Systems* 10 (2010), 345–379.

[7] Sarah Barrington and Hany Farid. 2023. A comparative analysis of human and AI performance in forensic estimation of physical attributes. *Scientific Reports* 13, 1 (2023), 4784.

[8] David Bau, Hendrik Strobelt, William Peebles, Jonas Wulff, Bolei Zhou, Jun-Yan Zhu, and Antonio Torralba. 2019. Semantic photo manipulation with a generative image prior. *ACM Transactions on Graphics* 38, 4 (Jul. 2019), 11 pages.

[9] Kratika Bhagtani, Amit Kumar Singh Yadav, Emily R. Bartusiak, Ziyue Xiang, Ruiting Shao, Sriram Baireddy, and Edward J. Delp. 2022. An overview of recent work in media forensics: Methods and threats. arXiv:2204.12067. Retrieved from https://arxiv.org/abs/2204.12067

[10] Mikołaj Bińkowski, Danica J. Sutherland, Michael Arbel, and Arthur Gretton. 2021. Demystifying MMD GANs. arXiv:1801.01401. Retrieved from https://arxiv.org/abs/1801.01401

[11] Matyáš Boháček and Hany Farid. 2023. A geometric and photometric exploration of GAN and diffusion synthesized faces. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 874–883.

[12] Ali Borji. 2023. Qualitative failures of image generation models and their application in detecting deepfakes. *Image and Vision Computing* 137 (2023), 104771.

[13] Andrew Brock, Jeff Donahue, and Karen Simonyan. 2018. Large scale GAN training for high fidelity natural image synthesis. arXiv:1809.11096. Retrieved from https://arxiv.org/abs/1809.11096

[14] Junyi Cao, Chao Ma, Taiping Yao, Shen Chen, Shouhong Ding, and Xiaokang Yang. 2022. End-to-end reconstruction-classification learning for face forgery detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 4113–4122.

[15] Qiong Cao, Li Shen, Weidi Xie, Omkar M. Parkhi, and Andrew Zisserman. 2018. VGGFace2: A dataset for recognising faces across pose and age. In *13th IEEE International Conference on Automatic Face & Gesture Recognition (FG)*, 67–74.

[16] Yihan Cao, Siyu Li, Yixin Liu, Zhiling Yan, Yutong Dai, Philip S. Yu, and Lichao Sun. 2023. A comprehensive survey of AI-generated content (AIGC): A history of generative AI from GAN to ChatGPT. arXiv:2303.04226. Retrieved from https://arxiv.org/abs/2303.04226

[17] Nicholas Carlini and Hany Farid. 2020. Evading deepfake-image detectors with white-and black-box attacks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2804–2813.

[18] Anpei Chen, Ruiyang Liu, Ling Xie, Zhang Chen, Hao Su, and Jingyi Yu. 2021. SofGAN: A portrait image generator with dynamic styling. *ACM Transactions on Graphics* 41, 1 (Feb. 2021), 26 pages.

[19] Beijing Chen, Xingwang Ju, Bin Xiao, Weiping Ding, Yuhui Zheng, and Victor Hugo C. de Albuquerque. 2021. Locally GAN-generated face detection based on an improved Xception. *Information Sciences* 572 (2021), 16–28.

[20] Beijing Chen, Xin Liu, Yuhui Zheng, Guoying Zhao, and Yun-Qing Shi. 2022. A robust GAN-generated face detection method based on dual-color spaces and an improved Xception. *IEEE Transactions on Circuits and Systems for Video Technology* 32, 6 (2022), 3527–3538.

[21] Beijing Chen, Weijin Tan, Yiting Wang, and Guoying Zhao. 2022. Distinguishing between natural and GAN-generated face images by combining global and local features. *Chinese Journal of Electronics* 31, 1 (2022), 59–67.

[22] Tiewen Chen, Shanmin Yang, Shu Hu, Zhenghan Fang, Ying Fu, Xi Wu, and Xin Wang. 2024. Masked conditional diffusion model for enhancing deepfake detection. In *International Joint Conference on Neural Networks (IJCNN)*. IEEE, 1–7.

[23] Bowen Cheng, Ross Girshick, Piotr Dollár, Alexander C. Berg, and Alexander Kirillov. 2021. Boundary IoU: Improving object-centric image segmentation evaluation. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 15329–15337.

[24] Harry Cheng, Yangyang Guo, Tianyi Wang, Liqiang Nie, and Mohan Kankanhalli. 2024. Diffusion facial forgery detection. arXiv:2401.15859. Retrieved from https://arxiv.org/abs/2401.15859

[25] Umur Aybars Ciftci, Ilke Demir, and Lijun Yin. 2020. FakeCatcher: Detection of synthetic portrait videos using biological signals. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024), 1.

[26] Marius Cordts, Mohamed Omran, Sebastian Ramos, Timo Rehfeld, Markus Enzweiler, Rodrigo Benenson, Uwe Franke, Stefan Roth, and Bernt Schiele. 2016. The cityscapes dataset for semantic urban scene understanding. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 3213–3223.

[27] Riccardo Corvi, Davide Cozzolino, Giada Zingarini, Giovanni Poggi, Koki Nagano, and Luisa Verdoliva. 2023. On the detection of synthetic images generated by diffusion models. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*.

[28] Naser Damer, Kiran Raja, Marius Süßmilch, Sushma Venkatesh, Fadi Boutros, Meiling Fang, Florian Kirchbuchner, Raghavendra Ramachandra, and Arjan Kuijper. 2021. ReGenMorph: Visibly realistic GAN generated face morphing attacks by attack re-generation. In *16th International Symposium on Advances in Visual Computing (ISVC '21)*. Springer-Verlag, Berlin, 251–264.

[29] L. Minh Dang, Syed Ibrahim Hassan, Suhyeon Im, Jaecheol Lee, Sujin Lee, and Hyeonjoon Moon. 2018. Deep learning based computer generated face identification using convolutional neural network. *Applied Sciences* 8, 12 (2018), 2610.

[30] Duc-Tien Dang-Nguyen, Cecilia Pasquini, Valentina Conotter, and Giulia Boato. 2015. RAISE: A raw images dataset for digital image forensics. In *Proceedings of the 6th ACM Multimedia Systems Conference*, 219–224.

[31] Defense Advanced Research Projects Agency (DARPA). 2024. Semantic Forensics (SemaFor). Retrieved from https://www.darpa.mil/research/programs/semantic-forensics

[32] Debayan Deb, Xiaoming Liu, and Anil K. Jain. 2023. Unified detection of digital and physical face attacks. In *IEEE 17th International Conference on Automatic Face and Gesture Recognition (FG)*, 1–8.

[33] Luca Debiasi, Naser Damer, Alexandra Moseguí Saladié, Christian Rathgeb, Ulrich Scherhag, Christoph Busch, Florian Kirchbuchner, and Andreas Uhl. 2019. On the detection of GAN-based face morphs using established morph detectors. In *Image Analysis and Processing (ICIAP '19)*. Elisa Ricci, Samuel Rota Bulò, Cees Snoek, Oswald Lanz, Stefano Messelodi, and Nicu Sebe (Eds.), 345–356.

[34] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. ImageNet: A large-scale hierarchical image database. In *IEEE Conference on Computer Vision and Pattern Recognition*, 248–255.

[35] Yufan Deng, Xin Deng, Yiping Duan, and Mai Xu. 2023. Diffusion-generated fake face detection by exploring wavelet domain forgery clues. In *International Conference on Wireless Communications and Signal Processing (WCSP)*. IEEE, Nanjing, China, 1–6.

[36] Nhu-Tai Do, In-Seop Na, and Soo-Hyung Kim. 2018. Forensics face detection from GANs using convolutional neural network. In *2018 International Symposium on Information Technology Convergence (ISITC)*, 376–379.

[37] Chengdong Dong, Ajay Kumar, and Eryun Liu. 2022. Think twice before detecting GAN-generated fake images from their spectral domain imprints. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 7855–7864.

[38] Wei Du and Shifei Ding. 2021. A survey on multi-agent deep reinforcement learning: From the perspective of challenges and applications. *Artificial Intelligence Review* 54, 5 (2021), 3215–3238.

[39] Ziv Epstein, Aaron Hertzmann, Memo Akten, Hany Farid, Jessica Fjeld, Morgan R. Frank, Matthew Groh, Laura Herman, Neil Leach, Robert Mahari, et al. 2023. Art and the science of generative AI. *Science* 380, 6650 (2023), 1110–1111.

[40] Bing Fan, Feng Ding, Guopu Zhu, Jiwu Huang, Sam Kwong, Pradeep K. Atrey, and Siwei Lyu. 2025. Generating higher-quality anti-forensics DeepFakes with adversarial sharpening mask. *ACM Transactions on Multimedia Computing, Communications, and Applications* (2025).

[41] Jianwei Fei, Yunshu Dai, Peipeng Yu, Tianrun Shen, Zhihua Xia, and Jian Weng. 2022. Learning second order local anomaly for general face forgery detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 20238–20248.

[42] Tao Fu, Ming Xia, and Gaobo Yang. 2022. Detecting GAN-generated face images via hybrid texture and sensor noise based features. *Multimedia Tools and Applications* 81, 18 (2022), 26345–26359.

[43] Yong Fu, Tanfeng Sun, Xinghao Jiang, Ke Xu, and Peisong He. 2019. Robust GANs-face detection based on dual-channel CNN network. In *12th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI)*, 1–5.

[44] Ying Fu, Xinyu Zhu, Xiaojie Li, Xin Wang, Xi Wu, Shu Hu, Yi Wu, Siwei Lyu, and Wei Liu. 2024. VB-KGN: Variational Bayesian kernel generation networks for motion image deblurring. *IEEE Transactions on Multimedia* 27 (2024), 2028–2042.

[45] Apurva Gandhi and Shomik Jain. 2020. Adversarial perturbations fool deepfake detectors. In *IEEE International Joint Conference on Neural Networks (IJCNN)*, 1–8.

[46] Candice R. Gerstner and Hany Farid. 2022. Detecting real-time deep-fake videos using active illumination. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 53–60.

[47] Michael Goebel, Lakshmanan Nataraj, Tejaswi Nanjundaswamy, Tajuddin Manhar Mohammed, Shivkumar Chandrasekaran, and B. S. Manjunath. 2020. Detection, attribution and localization of GAN generated images. arXiv:2007.10466. Retrieved from https://arxiv.org/abs/2007.10466

[48] Lore Goetschalckx, Alex Andonian, Aude Oliva, and Phillip Isola. 2019. GANalyze: Toward visual definitions of cognitive image properties. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 5743–5752.

[49] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, Vol. 27, 2672–2680.

[50] Jianping Gou, Baosheng Yu, Stephen J. Maybank, and Dacheng Tao. 2021. Knowledge distillation: A survey. *International Journal of Computer Vision* 129, 6 (2021), 1789–1819.

[51] Diego Gragnaniello, Davide Cozzolino, Francesco Marra, Giovanni Poggi, and Luisa Verdoliva. 2021. Are GAN generated images easy to detect? A critical analysis of the state-of-the-art. In *IEEE International Conference on Multimedia and Expo (ICME)*, 1–6.

[52] Qiqi Gu, Shen Chen, Taiping Yao, Yang Chen, Shouhong Ding, and Ran Yi. 2022. Exploiting fine-grained face forgery clues via progressive enhancement learning. In *AAAI Conference on Artificial Intelligence*, Vol. 36. AAAI Press, Palo Alto, California, 735–743.

[53] Hui Guo, Shu Hu, Xin Wang, Ming-Ching Chang, and Siwei Lyu. 2022. Eyes tell all: Irregular pupil shapes reveal GAN-generated faces. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2904–2908.

[54] Hui Guo, Shu Hu, Xin Wang, Ming-Ching Chang, and Siwei Lyu. 2022. Open-eye: An open platform to study human performance on identifying AI-synthesized faces. In *IEEE 5th International Conference on Multimedia Information Processing and Retrieval (MIPR)*, 224–227.

[55] Hui Guo, Shu Hu, Xin Wang, Ming-Ching Chang, and Siwei Lyu. 2022. Robust attentive deep neural network for exposing GAN-generated faces. *IEEE Access* 10 (2022), 32574–32583.

[56] Hui Guo, Xin Wang, and Siwei Lyu. 2023. Detection of real-time deepfakes in video conferencing with active probing and corneal reflection. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1–5.

[57] Travis Hartman and Raphael Satter. 2020. These Faces Are Not Real. Retrieved from https://tmsnrt.rs/3rsneCO

[58] Alexandru Hesson. 2023. *Detecting GAN-Generated Face Morphs Using Human Iris Characteristics*. B.S. thesis. University of Twente.

[59] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. 2017. GANs trained by a two time-scale update rule converge to a local nash equilibrium. In *31st International Conference on Neural Information Processing Systems (NIPS '17)*. Curran Associates Inc., Red Hook, NY, 6629–6640.

[60] Shu Hu, Lipeng Ke, Xin Wang, and Siwei Lyu. 2021. TkML-AP: Adversarial attacks to top-k multi-label learning. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 7649–7657.

[61] Shu Hu, Yuezun Li, and Siwei Lyu. 2021. Exposing GAN-generated faces using inconsistent corneal specular highlights. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2500–2504.

[62] Tai-Ming Huang, Yue Hua Han, Ernie Chu, Shu-Tzu Lo, Kai-Lung Hua, and Jun-Cheng Chen. 2025. Generalized image-based deepfake detection through foundation model adaptation. In *International Conference on Pattern Recognition*. Springer, 194–210.

[63] Nils Hulzebosch, Sarah Ibrahimi, and Marcel Worring. 2020. Detecting CNN-generated facial images in real-world scenarios. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 642–643.

[64] Abdul Jabbar, Xi Li, and Bourahla Omar. 2021. A survey on generative adversarial networks: Variants, applications, and training. *ACM Computing Surveys* 54, 8 (Oct. 2021), 49 pages.

[65] Ali Jahanian, Lucy Chai, and Phillip Isola. 2020. On the "steerability" of generative adversarial networks. In *International Conference on Learning Representations*.

[66] Hyeonseong Jeon, Youngoh Bang, Junyaup Kim, and Simon S. Woo. 2020. T-GD: Transferable GAN-generated images detection framework. In *37th International Conference on Machine Learning (ICML '20)*. JMLR.org, Vienna, Austria, Article 441.

[67] Yonghyun Jeong, Doyeon Kim, Youngmin Ro, and Jongwon Choi. 2022. FrePGAN: Robust deepfake detection using frequency-level perturbations. In *AAAI Conference on Artificial Intelligence*, Vol. 36. AAAI Press, Vancouver, BC, Canada, 1060–1068.

[68] Wei Jiang and Wentao Dong. 2022. Facke: A survey on generative models for face swapping. arXiv:2206.11203. Retrieved from https://arxiv.org/abs/2206.11203

[69] Micah K. Johnson and Hany Farid. 2008. Exposing digital forgeries through specular highlights on the eye. In *Information Hiding*. Teddy Furon, François Cayre, Gwenaël Doërr, and Patrick Bas (Eds.), Springer, Berlin, 311–325. DOI: https://doi.org/10.1007/978-3-540-77370-2_21

[70] Yan Ju, Shu Hu, Shan Jia, George H. Chen, and Siwei Lyu. 2024. Improving fairness in deepfake detection. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 4655–4665.

[71] Felix Juefei-Xu, Run Wang, Yihao Huang, Qing Guo, Lei Ma, and Yang Liu. 2022. Countering malicious DeepFakes: Survey, battleground, and horizon. *International Journal of Computer Vision* 130, 7 (2022), 1678–1734.

[72] Sarthak Kamat, Shruti Agarwal, Trevor Darrell, and Anna Rohrbach. 2023. Revisiting generalizability in deepfake detection: Improving metrics and stabilizing transfer. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 426–435.

[73] Amina Kammoun, Rim Slama, Hedi Tabia, Tarek Ouni, and Mohmed Abid. 2022. Generative adversarial networks for face generation: A survey. *ACM Computing Surveys* 55, 5, Article 94 (Dec. 2022), 37 pages.

[74] Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. 2018. Progressive growing of GANs for improved quality, stability, and variation. arXiv:1710.10196. Retrieved from https://arxiv.org/abs/1710.10196

[75] Tero Karras, Miika Aittala, Samuli Laine, Erik Härkönen, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. 2021. Alias-free generative adversarial networks. In *Advances in Neural Information Processing Systems*, Vol. 34. Curran Associates, Inc., Red Hook, NY, 852–863.

[76] Tero Karras, Samuli Laine, and Timo Aila. 2019. A style-based generator architecture for generative adversarial networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 4401–4410.

[77] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. 2020. Analyzing and improving the image quality of StyleGAN. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 8107–8116.

[78] Zahra Khanjani, Gabrielle Watson, and Vandana P. Janeja. 2021. How deep are the fakes? Focusing on audio deepfake: A survey. arXiv:2111.14203. Retrieved from https://arxiv.org/abs/2111.14203

[79] Ali Khodabakhsh, Raghavendra Ramachandra, and Christoph Busch. 2019. Subjective evaluation of media consumer vulnerability to fake audiovisual content. In *11th IEEE International Conference on Quality of Multimedia Experience (QoMEX)*, 1–6.

[80] Rajat Kulkarni, Rutik Gaikwad, Rudraksh Sugandhi, Pranjali Kulkarni, and Shivraj Kone. 2019. Survey on deep learning in music using GAN. *International Journal of Engineering Research & Technology* 8, 9 (2019), 646–648.

[81] Federica Lago, Cecilia Pasquini, Rainer Bohme, Helene Dumont, Valerie Goffaux, and Giulia Boato. 2022. More real than real: A study on human visual perception of synthetic faces. *IEEE Signal Processing Magazine* 39, 1 (2022), 109–116.

[82] Christian Ledig, Lucas Theis, Ferenc Huszár, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, Alykhan Tejani, Johannes Totz, Zehan Wang, et al. 2017. Photo-realistic single image super-resolution using a generative adversarial network. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, Honolulu, HI, 4681–4690.

[83] Chuqiao Li, Zhiwu Huang, Danda Pani Paudel, Yabin Wang, Mohamad Shahbazi, Xiaopeng Hong, and Luc Van Gool. 2023. A continual deepfake detection benchmark: Dataset, methods, and essentials. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 1339–1349.

[84] Haodong Li, Bin Li, Shunquan Tan, and Jiwu Huang. 2020. Identification of deep network generated images using disparities in color components. *Signal Processing* 174 (Sep. 2020), 107616.

[85] Lin Li, Lixin Qin, Zeguo Xu, Youbing Yin, Xin Wang, Bin Kong, Junjie Bai, Yi Lu, Zhenghan Fang, Qi Song, et al. 2020. Using artificial intelligence to detect COVID-19 and community-acquired pneumonia based on pulmonary CT: Evaluation of the diagnostic accuracy. *Radiology* 296, 2 (2020), E65–E71.

[86] Yidong Li, Wenhua Liu, Yi Jin, and Yuanzhouhan Cao. 2021. SPGAN: Face forgery using spoofing generative adversarial networks. *ACM Transactions on Multimedia Computing, Communications, and Applications* 17, 1s, Article 19 (Mar. 2021), 20 pages.

[87] Quanyu Liao, Yuezun Li, Xin Wang, Bin Kong, Bin Zhu, Siwei Lyu, Youbing Yin, Qi Song, and Xi Wu. 2021. Imperceptible adversarial examples for fake image detection. In *IEEE International Conference on Image Processing (ICIP)*, 3912–3916.
[88] Chenhao Lin, Jingyi Deng, Pengbin Hu, Chao Shen, Qian Wang, and Qi Li. 2024. Towards benchmarking and evaluating deepfake detection. arXiv:2203.02115. Retrieved from https://arxiv.org/abs/2203.02115
[89] Li Lin, Neeraj Gupta, Yue Zhang, Hainan Ren, Chun-Hao Liu, Feng Ding, Xin Wang, Xin Li, Luisa Verdoliva, and Shu Hu. 2024. Detecting multimedia generated by large AI models: A survey. arXiv:2402.00045. Retrieved from https://arxiv.org/abs/2402.00045
[90] Li Lin, Xinan He, Yan Ju, Xin Wang, Feng Ding, and Shu Hu. 2024. Preserving fairness generalization in deepfake detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 16815–16825.
[91] Li Lin, Santosh, Mingyang Wu, Xin Wang, and Shu Hu. 2025. AI-Face: A million-scale demographically annotated AI-generated face dataset and fairness benchmark. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
[92] Ping Liu, Qiqi Tao, and Joey Tianyi Zhou. 2025. Evolving from single-modal to multi-modal facial deepfake detection: Progress and challenges. arXiv:2406.06965. Retrieved from https://arxiv.org/abs/2406.06965
[93] Xin Liu and Xiao Chen. 2020. A survey of GAN-generated fake faces detection method based on deep learning. *Journal of Information Hiding and Privacy Protection* 2, 2 (2020), 87–94.
[94] Yunfan Liu, Qi Li, Qiyao Deng, Zhenan Sun, and Ming-Hsuan Yang. 2023. GAN-based facial attribute manipulation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 12 (2023), 14590–14610.
[95] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. 2015. Deep learning face attributes in the wild. In *IEEE International Conference on Computer Vision (ICCV)*, 3730–3738.
[96] Zhengzhe Liu, Xiaojuan Qi, and Philip H. S. Torr. 2020. Global texture enhancement for fake face detection in the wild. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 8057–8066.
[97] Peter Lorenz, Ricard L. Durall, and Janis Keuper. 2023. Detecting images generated by deep diffusion models using their local intrinsic dimensionality. In *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 448–459.
[98] Ruipeng Ma, Jinhao Duan, Fei Kong, Xiaoshuang Shi, and Kaidi Xu. 2023. Exposing the fake: Effective diffusion-generated images detection. arXiv:2307.06272. Retrieved from https://arxiv.org/abs/2307.06272
[99] Asad Malik, Minoru Kuribayashi, Sani M. Abdullahi, and Ahmad Neyaz Khan. 2022. DeepFake detection for human face images and videos: A survey. *IEEE Access* 10 (2022), 18757–18775.
[100] Hadi Mansourifar and Weidong Shi. 2020. One-shot GAN generated fake face detection. arXiv:2003.12244. Retrieved from https://arxiv.org/abs/2003.12244
[101] Francesco Marra, Diego Gragnaniello, Luisa Verdoliva, and Giovanni Poggi. 2019. Do GANs leave artificial fingerprints? In *IEEE Conference on Multimedia Information Processing and Retrieval (MIPR)*, 506–511.
[102] Francesco Marra, Cristiano Saltori, Giulia Boato, and Luisa Verdoliva. 2019. Incremental learning for the detection and classification of GAN-generated images. In *IEEE International Workshop on Information Forensics and Security (WIFS)*, 1–6.
[103] Momina Masood, Mariam Nawaz, Khalid Mahmood Malik, Ali Javed, Aun Irtaza, and Hafiz Malik. 2023. Deepfakes generation and detection: State-of-the-art, open challenges, countermeasures, and way forward. *Applied Intelligence* 53, 4 (2023), 3974–4026.
[104] Falko Matern, Christian Riess, and Marc Stamminger. 2019. Exploiting visual artifacts to expose deepfakes and face manipulations. In *IEEE Winter Applications of Computer Vision Workshops (WACVW)*, 83–92.
[105] Scott McCloskey and Michael Albright. 2018. Detecting GAN-generated imagery using color cues. arXiv:1812.08247. Retrieved from https://arxiv.org/abs/1812.08247
[106] Yisroel Mirsky and Wenke Lee. 2021. The creation and detection of deepfakes: A survey. *ACM Computing Surveys* 54, 1 (2021), 1–41.
[107] Huaxiao Mo, Bolin Chen, and Weiqi Luo. 2018. Fake faces identification via convolutional neural network. In *6th ACM Workshop on Information Hiding and Multimedia Security (IH&MMSec '18)*, 43–47.
[108] Shivansh Mundra, Gonzalo J. Aniano Porcile, Smit Marvaniya, James R. Verbus, and Hany Farid. 2023. Exposing GAN-generated profile photos from compact embeddings. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 884–892.
[109] Lakshmanan Nataraj, Tajuddin Manhar Mohammed, B. Manjunath, Shivkumar Chandrasekaran, Arjuna Flenner, Md Jawadul Bappy, and Amit Roy-Chowdhury. 2019. Detecting GAN-generated fake images using co-occurrence matrices. In *IS&T International Symposium on Electronic Imaging*, 532.
[110] Minh-Quang Nguyen, Khanh-Duy Ho, Hoang-Minh Nguyen, Canh-Minh Tu, Minh-Triet Tran, and Trong-Le Do. 2023. Unmasking the artist: Discriminating human-drawn and AI-generated human face art through facial feature analysis. In *International Conference on Multimedia Analysis and Pattern Recognition (MAPR)*, 1–6.

[111] Thanh Thi Nguyen, Quoc Viet Hung Nguyen, Dung Tien Nguyen, Duc Thanh Nguyen, Thien Huynh-The, Saeid Nahavandi, Thanh Tam Nguyen, Quoc-Viet Pham, and Cuong M. Nguyen. 2022. Deep learning for deepfakes creation and detection: A survey. *Computer Vision and Image Understanding* 223 (Oct. 2022), 103525.

[112] Yunsheng Ni, Depu Meng, Changqian Yu, Chengbin Quan, Dongchun Ren, and Youjian Zhao. 2022. CORE: Consistent representation learning for face forgery detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 12–21.

[113] Ahmad Nickabadi, Maryam Saeedi Fard, Nastaran Moradzadeh Farid, and Najmeh Mohammadbagheri. 2022. A comprehensive survey on semantic facial attribute editing using generative adversarial networks. arXiv:2205.10587. Retrieved from https://arxiv.org/abs/2205.10587

[114] Sophie Nightingale, Shruti Agarwal, Erik Härkönen, Jaakko Lehtinen, and Hany Farid. 2021. Synthetic faces: How perceptually convincing are they? *Journal of Vision* 21, 9 (2021), 2015.

[115] Sophie Nightingale and Hany Farid. 2022. Synthetic faces are more trustworthy than real faces. *Journal of Vision* 22, 14 (2022), 3068–3068.

[116] Sophie J. Nightingale, Shruti Agarwal, and Hany Farid. 2021. Perceptual and computational detection of face morphing. *Journal of Vision* 21, 3 (2021), 4.

[117] Sophie J. Nightingale and Hany Farid. 2022. AI-synthesized faces are indistinguishable from real faces and more trustworthy. *Proceedings of the National Academy of Sciences* 119, 8 (2022), e2120481119.

[118] Ehsan Nowroozi, Mauro Conti, and Yassine Mekdad. 2022. Detecting high-quality GAN-generated face images using neural networks. arXiv:2203.01716. Retrieved from https://arxiv.org/abs/2203.01716

[119] Kwang-Kyo Oh, Myoung-Chul Park, and Hyo-Sung Ahn. 2015. A survey of multi-agent formation control. *Automatica* 53 (2015), 424–440.

[120] Donie O'Sullivan. 2020. A High School Student Created a Fake 2020 Candidate. Twitter Verified It. CNN Business. Retrieved from https://cnn.it/3HpHfzz

[121] Donie O'Sullivan. 2020. How Fake Faces Are Being Weaponized Online. CNN Business. Retrieved from https://www.cnn.com/2020/02/20/tech/fake-faces-deepfake/index.html

[122] P. Swathi and Saritha Sk. 2021. DeepFake creation and detection: A survey. In *3rd International Conference on Inventive Research in Computing Applications (ICIRCA)*, 584–588.

[123] Lorenzo Papa, Lorenzo Faiella, Luca Corvitto, Luca Maiano, and Irene Amerini. 2023. On the use of stable diffusion for creating realistic faces: From generation to detection. In *11th International Workshop on Biometrics and Forensics (IWBF)*. IEEE, Barcelona, Spain, 1–6.

[124] Taesung Park, Ming-Yu Liu, Ting-Chun Wang, and Jun-Yan Zhu. 2019. GauGAN: Semantic image synthesis with spatially adaptive normalization. In *ACM SIGGRAPH 2019 Real-Time Live!*, Article 2, 1 pages.

[125] Samay Pashine, Sagar Mandiya, Praveen Gupta, and Rashid Sheikh. 2021. Deep fake detection: Survey of facial manipulation detection solutions. arXiv:2106.12605. Retrieved from https://arxiv.org/abs/2106.12605

[126] Leandro A. Passos, Danilo Jodas, Kelton A. P. Costa, Luis A. Souza Júnior, Douglas Rodrigues, Javier Del Ser, David Camacho, and João Paulo Papa. 2024. A review of deep learning-based approaches for deepfake content detection. *Expert Systems* 41, 8 (Feb. 2024), e13570.

[127] Gan Pei, Jiangning Zhang, Menghan Hu, Zhenyu Zhang, Chengjie Wang, Yunsheng Wu, Guangtao Zhai, Jian Yang, Chunhua Shen, and Dacheng Tao. 2024. Deepfake generation and detection: A benchmark and survey. arXiv:2403.17881. Retrieved from https://arxiv.org/abs/2403.17881

[128] Ivan Perov, Daiheng Gao, Nikolay Chervoniy, Kunlin Liu, Sugasa Marangonda, Chris Umé, Dpfks, Carl Shift Facenheim, Luis R. P., Jian Jiang, et al. 2021. DeepFaceLab: Integrated, flexible and extensible face-swapping framework. arXiv:2005.05535. Retrieved from https://arxiv.org/abs/2005.05535

[129] Minh Tam Pham, Thanh Trung Huynh, Thanh Tam Nguyen, Thanh Toan Nguyen, Thanh Thi Nguyen, Jun Jo, Hongzhi Yin, and Quoc Viet Hung Nguyen. 2024. A dual benchmarking study of facial forgery and facial forensics. arXiv:2111.12912. Retrieved from https://arxiv.org/abs/2111.12912

[130] Erion-Vasilis Pikoulis, Zafeiria-Marina Ioannou, Mersini Paschou, and Evangelos Sakkopoulos. 2021. Face morphing, a modern threat to border security: Recent advances and open challenges. *Applied Sciences* 11, 7 (2021), 3207.

[131] Gonzalo J. Aniano Porcile, Jack Gindi, Shivansh Mundra, James R. Verbus, and Hany Farid. 2024. Finding AI-generated faces in the wild. arXiv:2311.08577. Retrieved from https://arxiv.org/abs/2311.08577

[132] Wenbo Pu, Jing Hu, Xin Wang, Yuezun Li, Shu Hu, Bin Zhu, Rui Song, Qi Song, Xi Wu, Siwei Lyu, et al. 2022. Learning a deep dual-level network for robust DeepFake detection. *Pattern Recognition* 130 (2022), 108832.

[133] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning. PMLR*, 8748–8763.

[134] Alec Radford, Luke Metz, and Soumith Chintala. 2016. Unsupervised representation learning with deep convolutional generative adversarial networks. arXiv:1511.06434. Retrieved from https://arxiv.org/abs/1511.06434

[135] Omer Sagi and Lior Rokach. 2018. Ensemble learning: A survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 8 (2018), e1249.
[136] Takaya Saito and Marc Rehmsmeier. 2015. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS One* 10, 3 (2015), e0118432.
[137] Iman Saleem and Bahja Khudair Shukr. 2023. Techniques and challenges for generation and detection face morphing attacks: A survey. *Iraqi Journal of Science* 64, 1 (Jan. 2023), 385–404.
[138] Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. 2016. Improved techniques for training GANs. In *30th International Conference on Neural Information Processing Systems (NIPS '16)*, 2234–2242.
[139] Eklavya Sarkar, Pavel Korshunov, Laurent Colbois, and Sébastien Marcel. 2022. Are GAN-based morphs threatening face recognition? In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Singapore, 2959–2963.
[140] Ulrich Scherhag, Christian Rathgeb, Johannes Merkle, Ralph Breithaupt, and Christoph Busch. 2019. Face recognition systems under morphing attacks: A survey. *IEEE Access* 7 (2019), 23012–23026.
[141] Gil Shamai, Ron Slossberg, and Ron Kimmel. 2019. Synthesizing facial photometries and corresponding geometries using generative adversarial networks. *ACM Transactions on Multimedia Computing, Communications, and Applications* 15, 3s, Article 87 (Oct. 2019), 24 pages.
[142] Yujun Shen, Jinjin Gu, Xiaoou Tang, and Bolei Zhou. 2020. Interpreting the latent space of GANs for semantic face editing. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 9240–9249.
[143] Kaede Shiohara and Toshihiko Yamasaki. 2022. Detecting deepfakes with self-blended images. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 18699–18708.
[144] Karen Simonyan and Andrew Zisserman. 2015. Very deep convolutional networks for large-scale image recognition. arXiv:1409.1556. Retrieved from https://arxiv.org/abs/1409.1556
[145] Ruben Tolosana, Ruben Vera-Rodriguez, Julian Fierrez, Aythami Morales, and Javier Ortega-Garcia. 2020. Deepfakes and beyond: A survey of face manipulation and fake detection. *Information Fusion* 64 (2020), 131–148.
[146] Sushma Venkatesh, Raghavendra Ramachandra, Kiran Raja, and Christoph Busch. 2021. Face morphing attack generation and detection: A comprehensive survey. *IEEE Transactions on Technology and Society* 2, 3 (2021), 128–145.
[147] Sushma Venkatesh, Haoyu Zhang, Raghavendra Ramachandra, Kiran Raja, Naser Damer, and Christoph Busch. 2020. Can GAN generated morphs threaten face recognition systems equally as landmark based morphs? Vulnerability and detection. In *2020 8th International Workshop on Biometrics and Forensics (IWBF)*, 1–6.
[148] James Vincent. 2020. A Spy Reportedly Used an AI-Generated Profile Picture to Connect with Sources on LinkedIn. Retrieved from https://bit.ly/35BU215
[149] Jun Wang, Benedetta Tondi, and Mauro Barni. 2022. An eyes-based Siamese neural network for the detection of GAN-generated face images. *Frontiers in Signal Processing* 2 (2022), 918725.
[150] Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, et al. 2024. A survey on large language model based autonomous agents. *Frontiers of Computer Science* 18, 6 (2024), 186345.
[151] Run Wang, Felix Juefei-Xu, Lei Ma, Xiaofei Xie, Yihao Huang, Jian Wang, and Yang Liu. 2020. FakeSpotter: A simple yet robust baseline for spotting AI-synthesized fake faces. In *29th International Joint Conference on Artificial Intelligence (IJCAI '20)*. Christian Bessiere (Ed.), International Joint Conferences on Artificial Intelligence Organization, Yokohama, Japan, 3444–3451.
[152] Sheng-Yu Wang, Oliver Wang, Richard Zhang, Andrew Owens, and Alexei A. Efros. 2020. CNN-generated images are surprisingly easy to spot... for now. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 8692–8701.
[153] Xin Wang, Héctor Delgado, Hemlata Tak, Jee-Weon Jung, Hye-Jin Shim, Massimiliano Todisco, Ivan Kukanov, Xuechen Liu, Md Sahidullah, Tomi Kinnunen, et al. 2025. ASVspoof 5: Design, collection and validation of resources for spoofing, deepfake, and adversarial attack detection using crowdsourced speech. arXiv:2502.08857. Retrieved from https://arxiv.org/abs/2502.08857
[154] Xin Wang, Hui Guo, Shu Hu, Ming-Ching Chang, and Siwei Lyu. 2023. GAN-generated faces detection: A survey and new perspectives. In *European Conference on Artificial Intelligence*. IOS Press, Stockholm, Sweden, 2533–2542.
[155] Weihao Xia, Yulun Zhang, Yujiu Yang, Jing-Hao Xue, Bolei Zhou, and Ming-Hsuan Yang. 2021. GAN inversion: A survey. arXiv:2101.05278. Retrieved from https://arxiv.org/abs/2101.05278
[156] Junlin Xie, Zhihong Chen, Ruifei Zhang, Xiang Wan, and Guanbin Li. 2024. Large multimodal agents: A survey. arXiv:2402.15116. Retrieved from https://arxiv.org/abs/2402.15116
[157] Qijie Xu, Defang Chen, Jiawei Chen, Siwei Lyu, and Can Wang. 2025. Recent advances on generalizable diffusion-generated image detection. arXiv:2502.19716. Retrieved from https://arxiv.org/abs/2502.19716
[158] Ziyu Xue, Xiuhua Jiang, Qingtong Liu, and Zhaoshan Wei. 2023. Global-local facial fusion based GAN generated fake face detection. *Sensors* 23, 2 (2023), 616.

[159] Li-Chia Yang, Szu-Yu Chou, and Yi-Hsuan Yang. 2017. MidiNet: A convolutional generative adversarial network for symbolic-domain music generation. arXiv:1703.10847. Retrieved from https://arxiv.org/abs/1703.10847

[160] Rui Yang, Zhenrong Deng, Yushu Zhang, Xiaonan Luo, and Rushi Lan. 2024. 4DPM: Deepfake detection with a denoising diffusion probabilistic mask. *IEEE Signal Processing Letters* 31 (2024), 914–918.

[161] Shanmin Yang, Hui Guo, Shu Hu, Bin Zhu, Ying Fu, Siwei Lyu, Xi Wu, and Xin Wang. 2023. CrossDF: Improving cross-domain deepfake detection with deep information decomposition. arXiv:2310.00359. Retrieved from https://arxiv.org/abs/2310.00359

[162] Xin Yang, Yuezun Li, Honggang Qi, and Siwei Lyu. 2019. Exposing GAN-synthesized faces using landmark locations. In *ACM Workshop on Information Hiding and Multimedia Security (IH&MMSec '19)*, 113–118.

[163] Ning Yu, Larry Davis, and Mario Fritz. 2019. Attributing fake images to GANs: Learning and analyzing GAN fingerprints. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 7555–7565.

[164] Peipeng Yu, Zhihua Xia, Jianwei Fei, and Yujiang Lu. 2021. A survey on deepfake video detection. *IET Biometrics* 10, 6 (2021), 607–624.

[165] Yang Yu, Rongrong Ni, Wenjie Li, and Yao Zhao. 2022. Detection of AI-manipulated fake faces via mining generalized features. *ACM Transactions on Multimedia Computing, Communications, and Applications* 18, 4, Article 94 (Mar. 2022), 23 pages.

[166] Lei Zhang, Hao Chen, Shu Hu, Bin Zhu, Ching-Sheng Lin, Xi Wu, Jinrong Hu, and Xin Wang. 2024. X-Transfer: A transfer learning-based framework for GAN-generated fake image detection. In *IEEE International Joint Conference on Neural Networks (IJCNN)*, 1–8.

[167] Tao Zhang. 2022. Deepfake generation and detection, a survey. *Multimedia Tools and Applications* 81, 5 (2022), 6259–6276.

[168] Dragoş-Constantin Ţânţaru, Elisabeta Oneaţă, and Dan Oneaţă. 2024. Weakly-supervised deepfake localization in diffusion-generated images. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 6246–6256.

Received 30 August 2024; revised 3 April 2025; accepted 6 May 2025